

Warp knitting machinery

Three main types of warp knitting machines are

1. The tricot warp knitting machine
 - a) Mainly bearded and compound needle used for knitting
 - b) Finer gauge machine
2. The raschel warp knitting machine
 - a) latch needle used for knitting
 - b) Coarser gauge machine
3. The crochet warp knitting machine

Warp knitting machine according to number of needle bars :

1. Single needle bar warp knitting machine
 - a) Only plain knitted structures can be produced which is called single faced fabric.
 - b) Tricot machine (bearded needle)
 - c) Raschel machine (latch needle)
2. Double needle bar warp knitting machine
 - a) Rib knitted structures can be produced which is called double faced fabric.
 - b) Needle gaiting should be according to rib gaiting
 - c) Simplex machine (bearded needle)
 - d) Raschel machine (latch needle)

❖ According to the drive mechanism

1. Cams :

- a) In older machine
- b) Slow running machine reliable up to 500rpm

2. Eccentrics and crank system :

- a) Modern machines
- b) High speed machine
- c) Speeds up to 3500 rpm possible

❖ According to the function of sinkers used

a) Holding down and knock over sinker(tricot machine)

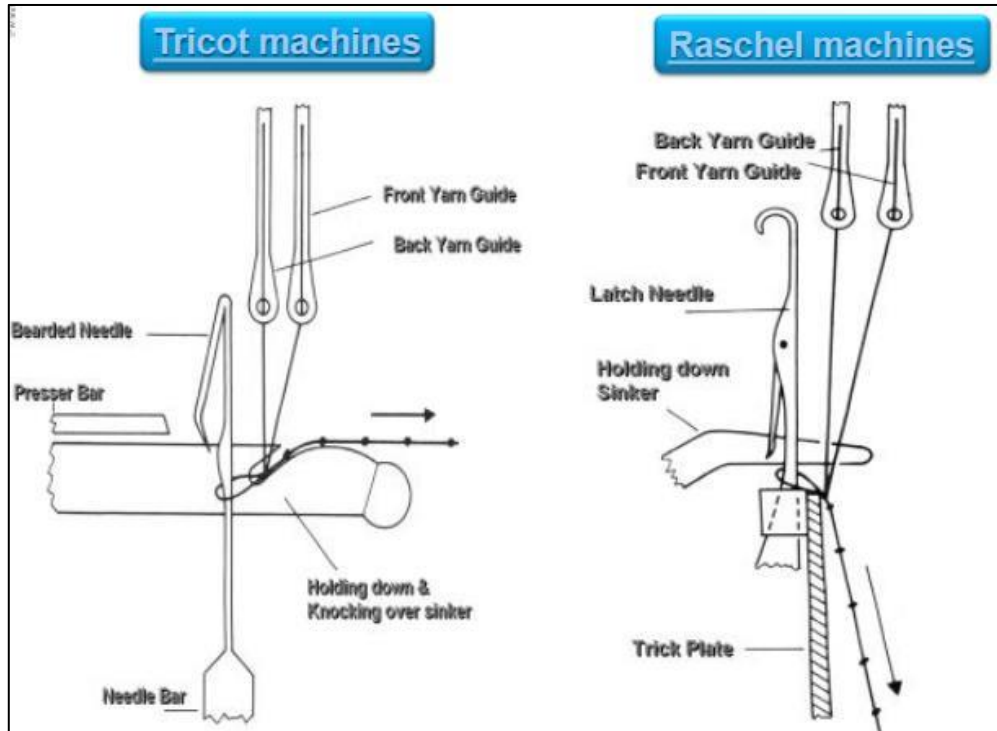
Holding down the knitted loops and provision of a platform for interconnection of knitted loops

b) Holding down sinker (raschel machine)

Holding down the knitted loops

❖ **According to the direction of fabric removal from the needles :**

- a) Tricot machine : at right angle to the needle stem
- b) Raschel machine : almost parallel to the needle stems (120 to 160 degree)



Basic construction of warp knitting machine :

- 1. Yarn supply system : warp beams where yarn is stored
- 2. Yarn delivery in form of a sheet of yarn
- 3. At least one yarn per needle
- 4. All knitting elements are mounted on bars and has collective movement
- 5. Vertical material flow

Warp beam let off mechanism

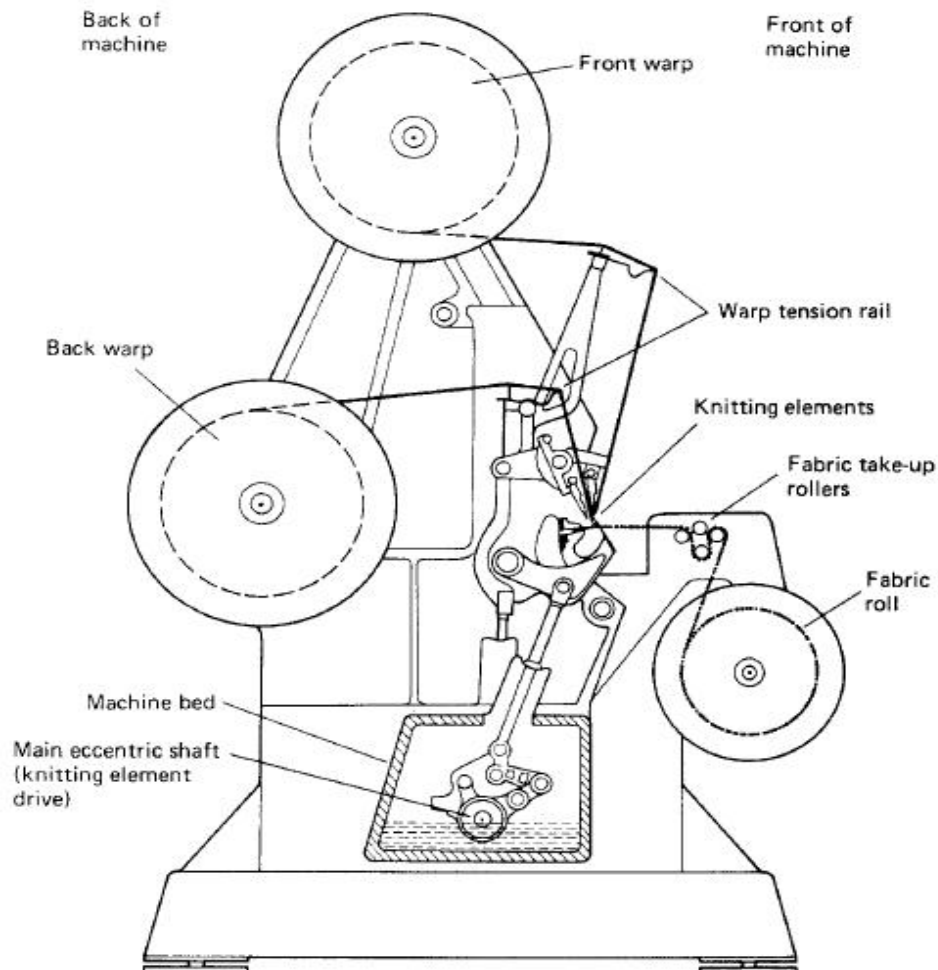
- 1. **Positive let off** : to deliver a per-determined rate of feed of the warp yarn to the needles
- 2. **Negative let off** : tension on the warp beam causes the warp yarn to be pulled from the beam as it rotates against a control friction .

Both must deliver yarn to the needles at the correct speed and tension ,failure to achieve this will causes yarn breakage or faults in the fabric .Incorrect setting of the let off will also changes the shape of stitch altering the fabric quality

Fabric take up or take down mechanism :

- Raschel pulls the fabric parallel to the needles
- This enable a high take up tension to operate which is ideal for coarser gauge open work,lace,net .
- Tricot pulls the fabric at right angle to the needles
- This enables lower take off tension which is ideal for finer gauge single needle bar pattern work
- The fabric take up is a contributing factor to the quality of the knitted fabric

Tricot warp knitting machine



Features of tricot warp knitting machine

- Type of needle: Tricot m/cs mainly employed bearded needles with a pressure bar. But now, almost all available commercial tricot fabric produced by compound needle.
- Machine gauge :Tricot m/cs have a gauge expressed in needle per inch
- Chain link numbering: chain link numbering 0, 1, 2, 3, 4 etc. generally with three links per course.
- Tricot sinker: Their sinkers, which are joined to each other at the front and back, never move clear of the needles as they combine the functions of holding down, knocking over and supporting the fabric
- The Fabric is drawn-away towards the batching roller almost at right angles to the needles bar.
- The warp beams are accommodated in an inclined arc towards the back of the m/c with the top beam supplying the front guide bar and the bottom beam supplying the back guide bar
- The warp sheets pass over the top of the guide bar rocker shaft to their tension rails situated at the front of the m/c
- Mechanical attention to the knitting elements is carried out at the front of the m/c, as the beams prevent access to the back.
- As all the warp sheets are drawn over the rocker shaft to the m/c, it is easier to thread up the guide bars commencing with the needle bar.
- The guide bars are therefore numbered from the back towards the front of the m/c because of threading sequence.

Function of different parts of M/C:

1. **Compound needle:** In Tricot warp knitting m/c compound needle is used. To form loop and produce the fabric is the main function of the needle.
2. **Needle bar:** A needle bar is used in this m/c. The main function of it is to hold the needles together and helps the needles to move unison while loop forming.
3. **Guide bar:** Guide bar is used in this m/c to guide the yarn properly to the needle. It feeds the yarn around the needle and controls the rate of warp feed from the warp beam by making lapping movement.
4. **Sinker and Sinker bar:** In the tricot warp knitting m/c tricot sinker is used to hold down the loops produced by the needles. The sinker bar keeps the sinkers together to move unison while knitting. The main function of the sinker is to hold down, knock over and supporting the fabric loops.

5. Sliding latch: In warp knitting m/c compound needles are used. A sliding latch is used here to close the hook while knitting.

6. Sliding latch bar: In warp knitting m/c the sliding latches remain unison. The latch bar keeps the latches together and helps it to move unison while knitting.

7. Cloth roller: The produced fabric is wound on the cloth roller.

8. Let-off mechanism: The process of releasing the warp yarns according to the requirement of the m/c and speed of cloth roller is the main function of let-off mechanism.

9. Take up mechanism: It helps the produced fabric to wound on the cloth roller in proper tension. There are 3 take-up r/r in this m/c. It also gives proper tension to the warp sheet and controls the speed of warp beam.

11. Link: In the warp knitting m/c the link is used to make design in the knitted fabric. The different links used here has different thickness and thus it helps to produce design.

12. Pattern chain: The pattern chain is the chain of links joined with each other. The pattern chain helps the m/c to produce design.

13. Pattern drum: It is a drum, which gives motion to the pattern chain. There are groove on it and the pattern chain is placed on it. It gets motion from the m/c driving motor through gearing.

14. Comb: In this warp knitting m/c the comb is used to separate the warp yarns coming from the warp beam. It works as the reeds of the weaving m/c and also controls the fabric width.

15. Warp beam: In tricot warp knitting m/c warp beam is used to supply the warp yarns parallelly. There are 8-warp beam in this m/c. 4 beam are on the upper side and 4 are to the downside. These beams are not so big as the weavers beam.

16. Machine A/C: The m/c a/c is one kind of cooling device, which keeps the m/c parts and the motor from over heating and damage while running continuously.

17. Main shaft: The main shaft of this m/c gives the m/c motion from the motor.

18. Lubrication: The m/c has automatic lubrication system.

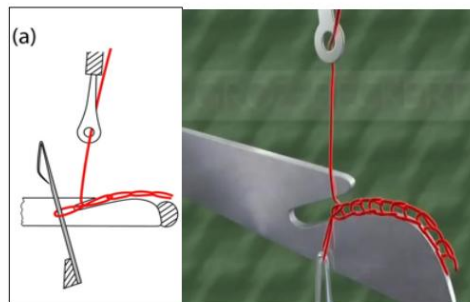
19. Controlling unit: The controlling unit is used to control the m/c easily. It has a digital control unit.

20. Toothed belt: In this m/c the toothed belts are used to transfer motion without any slippage.

Knitting cycle of bearded needle tricot warp machine

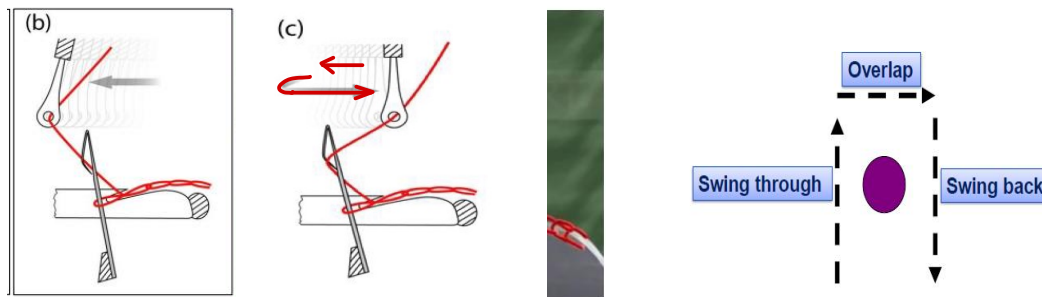
1. *The rest position (a):*

The needles have risen to $\frac{2}{3}$ of their full height from knock-over and have their beards towards the back of the machine.



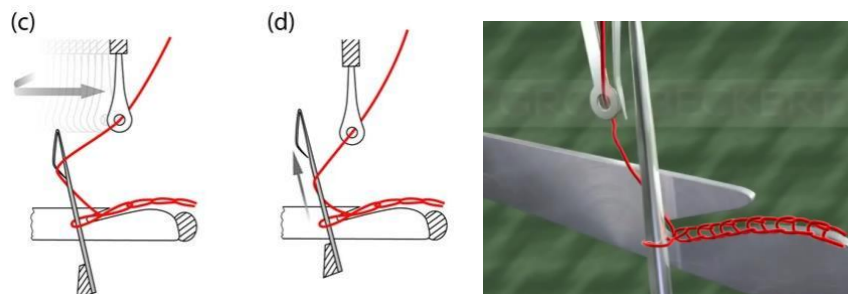
2. *Backward swing and overlap shog (b, c).*

The guide bar swings to the beard side of the needle and moving laterally towards the observer which is called shog and wraps the yarn around the beard of the needle.



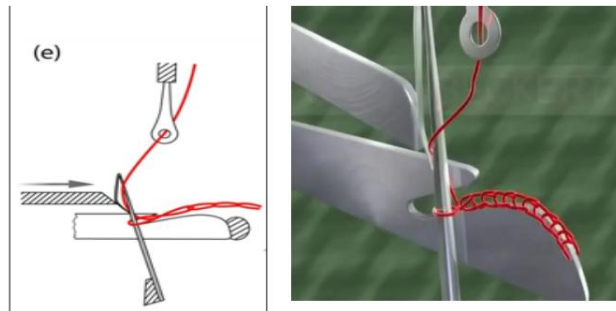
3. *The return swing and second rise (c, d).*

As the guides swing to the front, the needles rise to their full height so that the newly-formed overlaps placed onto the stems above the old overlaps.



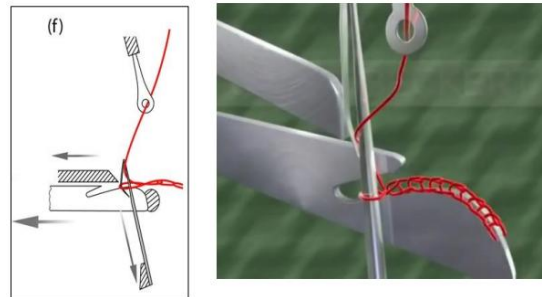
4. *Pressing (e).*

The needle bar descends so that the presser can press the beard to close the hook.



5. *Landing (f).*

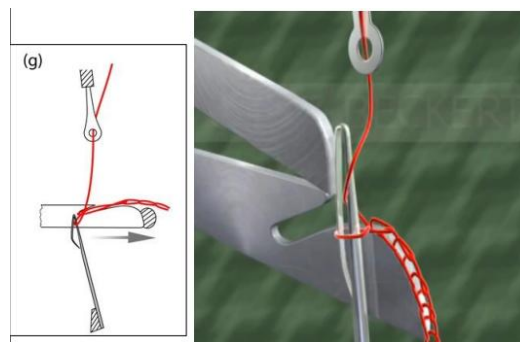
As the sinkers withdraw, the upward curve of their bellies lands the old overlaps onto the closed beards.



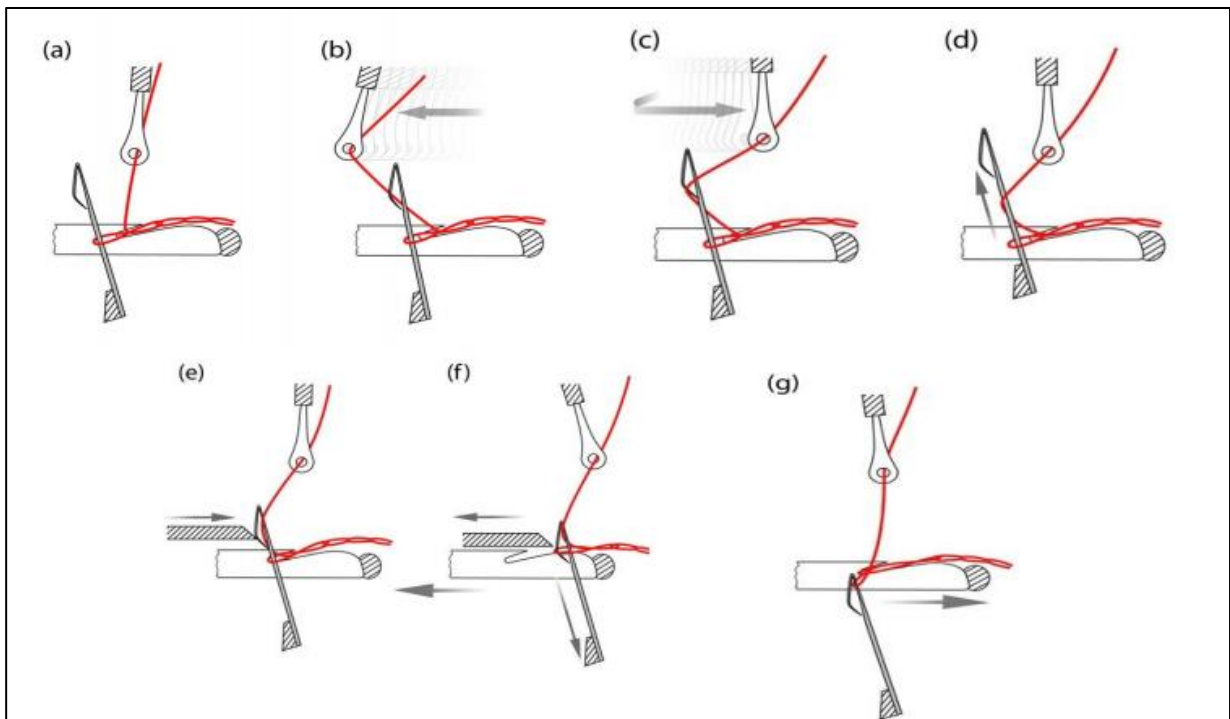
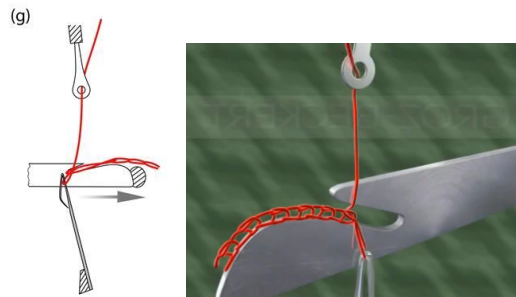
6. *Knock-over and underlap shog (g)*

The needles continue to descend while the sinker shift forward. The old and new loops enter the throat of sinker.

The underlap shog which can occur at any time between pressing and knock-over usually occurs in opposite directions on the two guide bars.



7. *The sinkers now move forward* to hold down the fabric loops and push them away from the ascending needles, which are rising to the rest position.



Knitting action of single needle bar latch compound needle Raschel warp knitting machine :

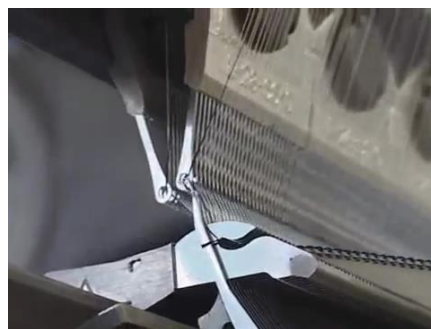
Step : 1 Needle rises & guide bar swing:

The needle rises until it reaches to the hole of the guide. Then the guide swing through to the back of the machine.



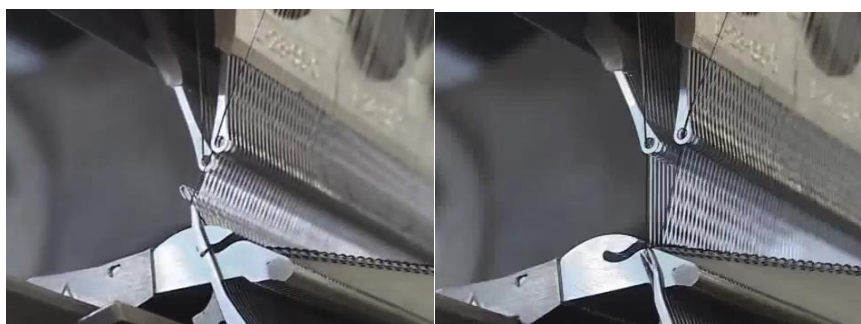
Step 2: The overlap & return swing :

The guide shog for overlap and swing to the front of the machine

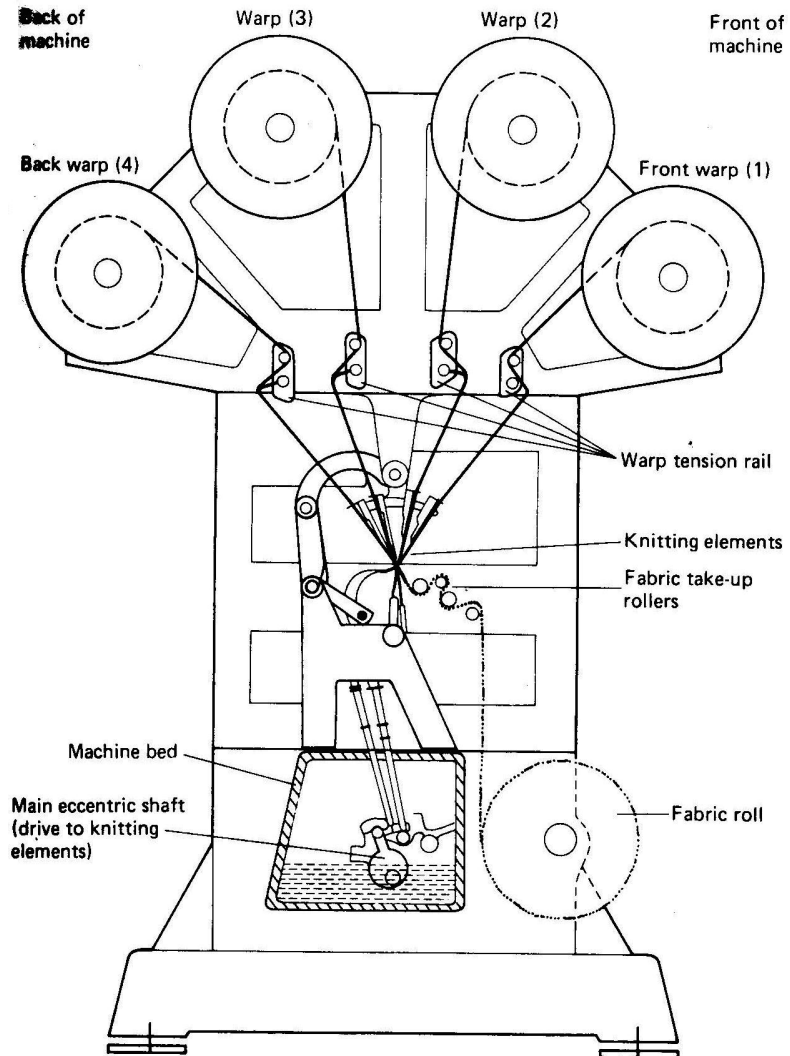


Step 03: Landing & knock over

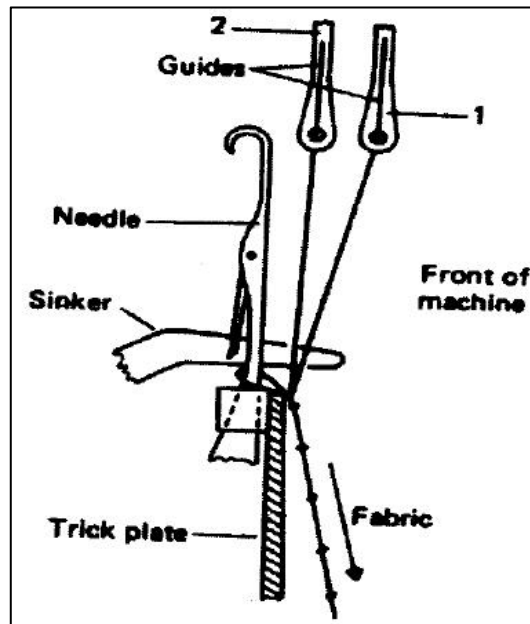
As the needle descend , the sinker start to withdraw so that the old loop can land onto the closed hook and then knock over as it descend below the sinker belly. At this moment under lap occurs before the needle begin their upward rises and sinker move forward to hold down the fabric.



Raschel machine



Function of different parts of Raschel warp knitting machine :



1 . Needle:

Most of the modern Rachel knitting machines built today use compound needle. Many m/cs, however, are still equipped with latch needles. Compound needles are set as in Tricot m/cs into trick which are cut into the needle bar and both main parts and closing elements are driven separately to open and close the hook.

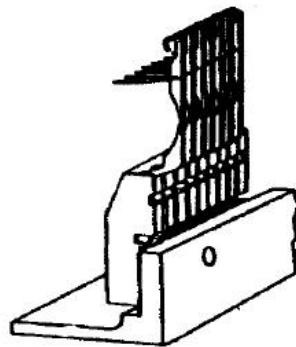


Fig: Latch needle unit

The latch needles, especially developed for those machines, are cast in units, one inch long. The latch of the needle depends for its [knitting operation](#), on the yarn.

2. Trick plate:

The loop formation in a Rachel m/c takes place on the upper edge of the trick plate. This bar can be considered as a needle bed, since the needles are placed in tricks cut into it.

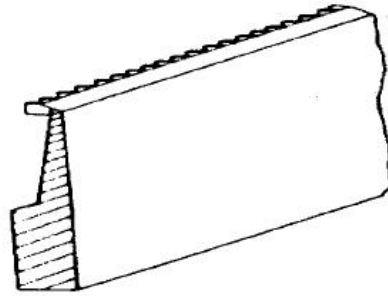
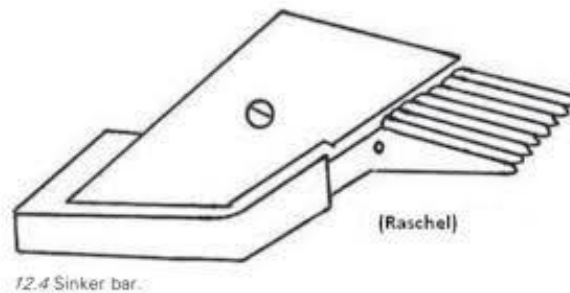


Fig: A section of the Trick Plate

3. Sinker:

The sinker like the latch needles are cast in units 1" long placed in the spaces between the needles. They are used to ensure that the fabric stays bowl while the needle rise to form their next knitting cycles. The sinker of Rachel m/cs are therefore only of secondary importance.



4. Latch guard:

A steel wire which are extended from one ends to the others parallel to the needle. During loop formation, the fabric loops open needle latch but sometimes closed the needle hook. As a result a closed hook cannot feed the new yarn. This latch guard placed needle hook side. So that the latch cannot close the hook and helps to open the needle hook.

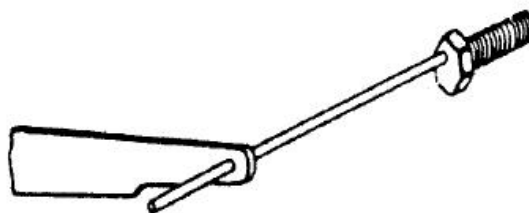


Fig: Latch guide

5. Guide & Guide bar:

The number of guide bar in Rachel m/c is more. Sometimes are of more than 70. Two types of guide bar are used-

- **Fully threaded guide bar:** Used to produce ground fabric and yarn pass to each guide.

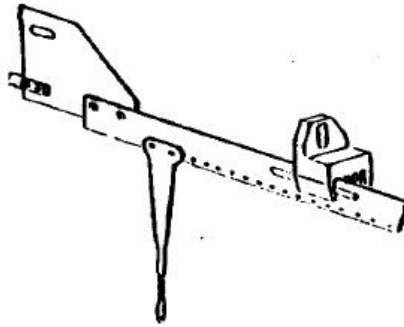


Fig: Guide bar with a guide finger

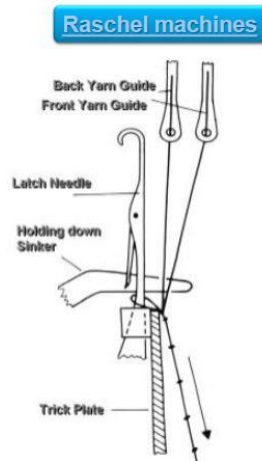
- **Partially threaded guide bar:** According to the design. One yarn is passed in each bar for each pattern repeat. Thread pass where design placed.

Features of Raschel warp knitting machine

1. Rachel m/cs used latch needles together with a wire or blade. Presently compound needle are also used at Raschel machine.
2. It has a gauge expressed in needles per 2inches (5cm) so that for example a 36 gauge Rachel m/c will have 18npi. There is a wide gauge ranges from E1 to E 32.
3. Their chain links are usually numbered in even numbers 0, 2, 4,etc. generally with 2 links/course.
4. Rachel sinkers only performs the function of holding down the loops whilst the needles rise
5. Rachel sinkers are not joined together by a lead across their ends nearest to the needle bars so they can move away clear of the needles towards the back of the m/c for the rest of the knitting cycle

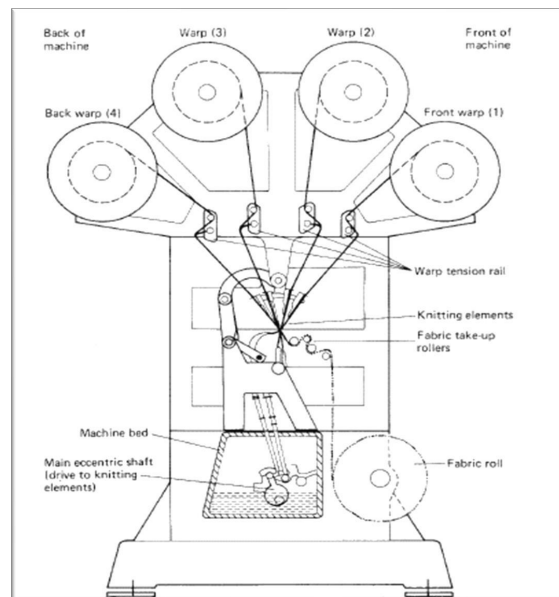


6. Their needle trick verge acts as a fabric support ledge and knock over surface.



7. The fabric is drawn downwards from the needles, almost parallel to the needle bar at an angle of 120° - 160° by a series of take-down rollers. This creates a high take up tension, particularly suitable for open fabric structure such as laces and nets.

8. The warp beams are arranged above the needle bar, centered over the rocker shaft, so that the warp sheets pass down to the guide bars on either side of it



9. The beams are placed above the m/cs so that it is accessible at the front for the fabric inspection

10. Mechanical attention are carried out at the back of the machine

11. The guide bars are threaded, connecting with the middle bars and working outwards from the either side of the rocker shaft

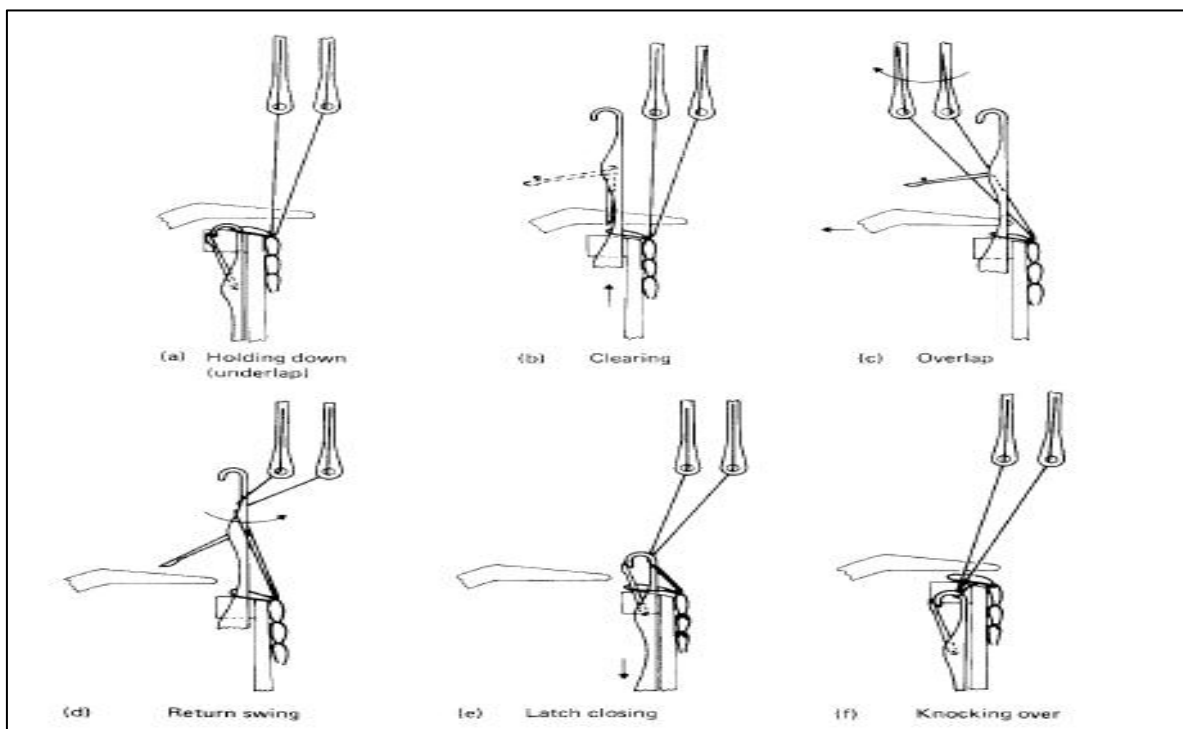
12. The guide bars are numbered from the front of the m/c

13. There is accommodation for at least four 32-inch diameter beams or large numbers of small diameter pattern bars

Knitting action of single needle bar latch needle Raschel warp

knitting machine(Knitting action of single needle bar latch needle Raschel warp knitting machine)

Raschel needles tend to have longer latches than weft knitting machine needles, to ensure that the wrapped yarns of the overlap goes onto and not below the open latch (Fig. 2.6). There is a trick-plate extending the full width of the machine, whose walls preserve the needle spacing and whose verge provides an edge for a clean knock-over. Holding-down sinkers that are thin blades, unled at their forward edges, move in a horizontal plain over the top of the trick-plate.



1. Holding down. The guide bars are at the front of the machine, completing their underlap shog. The sinker bar moves forward to hold the fabric down whilst the needle bar starts to rise from knock-over.

2. Clearing. As the needle bar rises to its full height, the old overlaps slip down onto the stems after opening the latches, which are prevented from flicking closed by latch wires. The sinker bar then starts to withdraw to allow the guide bars to overlap

3. Overlap. The guide bars swing to the back of the machine and then shog for the overlap.

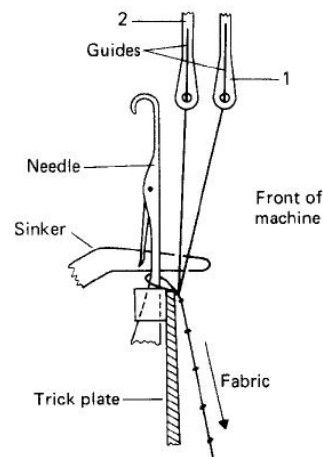
4. Return swing. As the guide bars swing to the front, the warp threads wrap into the needle hooks.

5. Latch closing. The needle bar descends so that the old overlaps contact and close the latches, trapping the new overlaps inside. The sinker bar now starts to move forward.

6. knocking-over and underlap. As the needle bar continues to descend, its head passes below the surface of the trick-plate, drawing the new overlap through the old overlap which is cast-off and as the sinkers advance over the trick-plate, the underlap shog of the guide bar is commenced.

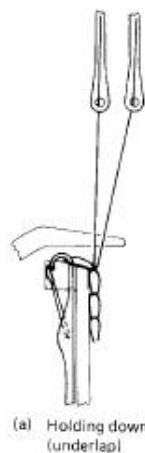
Knitting action of single needle bar latch needle Raschel warp knitting machine

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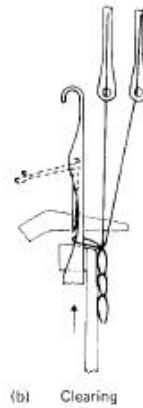
1. Holding down.

- The guide bars are at the front of the machine, completing their underlap shog.
- The sinker bar moves forward to hold the fabric down whilst the needle bar starts to rise from knock-over.



2. *Clearing.*

- As the needle bar rises to its full height, the old overlaps slip down onto the stems after opening the latches, which are prevented from flicking closed by latch wires
- The sinker bar then starts to withdraw to allow the guide bars to overlap



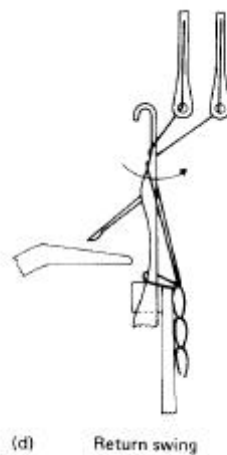
Overlap:

The **guide** bars swing to the back of the machine and then shog for the overlap.



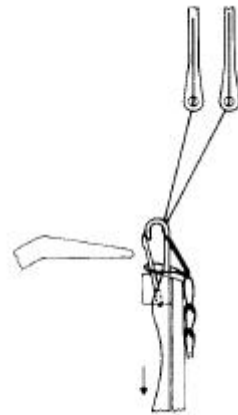
3. *Return swing.*

As the guide bars swing to the front, the warp threads wrap into the needle hooks.



5. Latch closing.

- *The needle bar descends so that the old overlaps contact and close the latches, trapping the new overlaps inside.*
- The sinker bar now starts to move forward.



(e) Latch closing

4. knocking-over and underlap



(f) Knocking over

Difference between Tricot and Raschel machine

Parameter	Tricot	Raschel
1.Needle used	Spring bearded needle and compound needle mainly used	Mainly latch needle used for knitting
2.Gauge system	Machine gauge expressed in needle per inch	Machine gauge expressed in needle per 2 inch
3.Chain link numbering	Chain link numbering 0, 1, 2, 3,etc	Chain link numbered in even numbers 0, 2, 4,etc
4. Link per course	three	Two
5. Function of sinker	Function of sinker is holding down, knocking over and supporting the fabric	Sinker performs the function of holding down the loops while the needle rise
6. take off mechanism	The fabric is drawn away towards the batching roller almost at right angles to the needle bar	The fabric is drawn downwards from the needles almost parallel to the bar at an angle of 120°-160°
7. Beam arrangement	The warp beams are accommodated in an inclined arc towards the back of the m/c	The warp beams are arranged above the needle bar centered over the rocker shaft
8. Maintenance	Mechanical attention is carried out at the front of the m/c	Mechanical attention is carried out at the back of the m/c
9.passage of warp sheet	The warp sheets pass over the top of the guide bar rockershaft to the machine	The warp sheets pass down the guide bar
10. Guide bar numbering	The guide bars are numbered from the back towards the front of the m/c	The guide bars are numbered from front of the m/c
11. Production	High speed production	Low speed production